

Section B

Refer to the separate Resource Folder to answer questions **7**, **8** and **9**.

7. X-rays are commonly used to diagnose illness.

(a) Refer to **Figure 2** to answer the following questions.

(i) State the range of wavelength for X-rays. m [1]

(ii) Estimate the highest frequency of X-rays. Hz [1]

(b) Refer to **Figure 3** to answer the following questions.

(i) A patient has received a total radiation dose of 70 units from X-rays during the year.
Calculate how many chest X-rays this dose is equivalent to. [2]

number of chest X-rays =

(ii) A patient has an X-ray taken of their spine. Calculate the fraction of the yearly background radiation this dose is equivalent to.
(1 year = 365 days). [2]

fraction =

